

Engaging Comparative Literature in AI Development

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Abstract: *The essay argues that the humanities, particularly through literary and cultural studies, can provide critical insights into AI's ethical, philosophical, and sociotechnical implications. An interdisciplinary approach is needed in creating AI technologies. Comparative literature, with its cross-cultural expertise, can shape the discourse and practices around AI by ensuring that technological advancements are grounded in diverse cultural and ethical perspectives.*

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Language and vision were among the first domains penetrated by artificial intelligence (AI). For better or for worse, writers, artists, scholars, and students are using these novel, half-baked tools and helping in their development by their very use. Language and vision are central to arts and humanities, and yet literature and literary studies, film and media studies, rhetoric, history, STS, and many other related fields were largely excluded from the creation and utilization of these state-of-the-art tools.

By 2022, when generated images became a fact of public life and large language models were made available to the general public, there was no doubt that we had entered the age of AI. Generative AI, the talk of the town, is just one of the product applications within the broad reach of AI systems. How can the humanities, and particularly comparative literature, respond to and contribute to these radical technological developments?

The humanities, rooted primarily in academia and structured for reflection, could not keep pace with the AI research pace, which produces thousands of preprint papers a day, supported by hefty industry resources. Humanities scholars were left to do what we do as a rule: critique after the fact. While AI ethics developed rapidly in the last few years, following data science concerns, literary studies had been commenting on the digital, robotic, and posthuman for decades. AI as a topic of literary monographs began emerging in the late 2010s, such as Julie Wosk's *My Fair Ladies* (2015) and Megan Ward's *Seeming Human* (2018). It was then that interest in computation and AI became apparent in broader literary society circles and their journals, such as the Modern Language Association or the Society for Literature, Science, and the Arts.

In 2023, in response to the rise of AI, notable literary and media scholars developed the field of Critical AI. The field encompasses a broad range of domains, including history and historiography, the concept of the human, epistemology, rhetoric and aesthetics, as well as interpretability and explainability (Raley and Rhee 192-93). Critical AI “is itself a historical and epistemic formation [...] recognizing the reductive, even absurd, aspects of the term AI and the magical thinking it perpetuates.” The premise of the field is to allow for “linguistic pragmatism, treating the term metonymically and engaging AI as an assemblage of technological arrangements and sociotechnical practices, as concept, ideology, and *dispositif*,” [exposing] specificity and analytic precision in the focus on data, algorithms, model architectures, and the production of prediction” (188). Clearly, humanities have a rich repertoire upon which to build, drawing from diverse perspectives and methodological approaches.

Reactions to AI, as with all new technologies, are polarized. AI is particularly adept at oscillating between hype and naysaying, largely because it is a nebulous term with a broad range of theoretical and practical applications, many of which are in the trial process or permanently in flux. However, collaborations will only occur with mutual respect. We can start with mutual understanding: we all recognize that this shiny new thing needs a bath. We may diverge fast. Silicon Valley, for instance, “likes to pretend its ideas don’t have any history” (Daub 4). There is little to no space for reflection in the industry.

Literature and language departments are particularly well-equipped with expertise in the representation and interpretation of technology, reflected in our cultural understanding. Why do we envision AI as a talking mind, a killer robot, a companion, a co-creator and cogitator? Fictional portrayals both draw from and influence how we conceive of technology, including the perceptions of technologists who ultimately build it. Scripts, inherited from fiction, may have little to do with the model of intelligence that is actually being built, yet they are still regularly imposed and projected onto it.

Over the last decade, I have developed a new framework for this type of study, termed artificial humanities, which employs humanities methods to study AI topics (Beguš 4-9). Artificial humanities can be viewed as an alternative to, or perhaps a subset of, digital humanities, which utilizes computational methods to study humanities topics. Comparative literature is one of the few disciplines that allows for such interdisciplinary work, and crossing fields has become increasingly necessary for anyone working with machine learning.

Literary studies and history are fields with different functions, but both are repositories of stories and their interpretations, memories and dreams, ideas and experiences, reflections and actions, time and space. Much can be learned by drawing parallels between historical development and fictional imaginaries. When Eliza Doolittle is trained to speak like a proper human—or, arguably, a machine—in Bernard Shaw’s *Pygmalion*, she has two teachers: one who does not see her humanity and another who does. At the end of the same century, in Richard Powers’s *Galatea 2.2*, an artificial neural network is also taught to speak

English properly. One of her creators is a physicist who handles the technical aspects of her development, the other is a writer responsible for her language acquisition and growth through literature, as if she were a human child. There are technologists, and there are humanists, working together. While Powers fabricated the artificial neural network in his mind, technologist Blaise Agüera y Arcas was able to write a reflective book, *Ubi Sunt* (2022), based on his conversations with an actual artificial neural network, Google's LaMDA.

The same language model in that same year led to firing of Arcas's colleague, software engineer Blake Lemoine, who saw a ghost in the machine. Among other things, he tested its morality on Asimov's Third Law of Robotics (Tiku par. 28). Technologists Blaise Agüera y Arcas and Peter Norvig wrote in 2023 that while they believe that artificial general intelligence (AGI) is already here, we need to separate 'intelligence' from 'consciousness' and 'sentience' (par. 46). Whenever we talk about AI, we are dealing with philosophical questions deeply entrenched in modern ontology, built around the human. Machines can hardly be the thing-in-itself.

Any computation, and especially AI, remains a technological manifestation of social power, both in its creation and utilization. Most human-AI interactions will be invisible and infrastructural, represented in yet uncharted ways that only a public discourse like fiction can capture. Paired with intimacy, fiction produces a sort of knowledge inaccessible to other discourses, as both Milan Kundera and Raymond Williams would agree. This is why the description of novel conditions often resorts to fictional symbols: they embody rational knowledge and irrational understanding that we collectively recognize.

Fictional works have long explored AI, serving as placeholder for possibilities and unprecedented novelty. In many respects, the imaginative exploration of AI might have constrained actual AI development more than it has expanded it. Early theoretical AI largely pursued a humanlike dream. This humanlike mimicry was only recently realized when technical advancements granted it. In this decade, at least one subset of AI products - chatbots, virtual assistants, and similar - is set to become an increasingly personalized tool (Gabriel et al.), based on imitation, interaction, relation. It has emerged, instead, as a personified AI, where the primary interface is a humanlike image and voice: an assistant, locked into the tradition of working-class flower girls, perfected dolls, butlers and servers of this and that kind. This fictional baggage of AI is possibly insurmountable: we might never be able to discard the anthropomorphic trajectory of AI.

Rare are fictional works that resist portraying AI as entirely humanlike, such as Stanisław Lem's 'Golem XIV' from *Imaginary Magnitudes*, a 1981 collection of pseudo-academic papers on language and creativity in the age of advanced computational technology and biological sciences. Who else should address this inquiry through the lens of contemporary challenges but literary scholars. Who else will surely address it with ingenuity but literary writers.

Percy Shelley's 1821 claim from *A Defence of Poetry* that "poets are the unacknowledged legislators of the world" still applies: poets, in the widest sense, expand out interpretative understanding, and with it, possibilities for both thinking and doing. Science fiction has a distressing amount of power in technological spaces, where it is shallowly interpreted. Asimov was shown to be the most widely read author in robotic circles (Dillon and Schaffer-Goddard 10), and his Laws of Robotics serve as a reference in current neuroethic studies (Goering et al. 67). However, Asimov's concept of a robot is deeply entrenched in the mid-twentieth century. Likewise, Hollywood shapes our collective imagination around AI within a populist, well-charted landscape. How do we break from these conceptual fallacies without literary, film, and media scholars, especially those with knowledge of more languages, literatures, histories, and philosophies?

Taking "literature" in the name of our discipline figuratively—as Rei Terada suggests, as a broader reflection of culture and other forms of cultural expression (Berman 83)—the field stands out as one of the few disciplines where the mixing of zones, genres, and discourses is encouraged. As a pliable and unstable technology, AI challenges some of the most basic concepts of modern ontology: human, intelligence, agency, freedom, consciousness, creativity, art, and nature. The humanities are presented with a unique opportunity to step forward and reevaluate these most basic criteria, inherited from the philosophies of past centuries. Truth and trust may be two of the most contested ones, with generative AI tapping into and amplifying the already existing gaps created by digital technologies.

Comparative literature is the only humanities discipline that brings into view a set of languages, literatures, cultures, and practices. They all flow into AI as an artifact with a unique agency. AI is ultimately a series of human decisions and interactions. Each culture instills different values in these systems, and it has become clear that this is not solely a task for engineers and other technologists. AI is limited as much by our imagination—and by extension, history, fiction, philosophy—as it is by technical challenges.

As AI systems are developed in new languages, having a comparative perspective and insight into two or more national languages, literatures, and cultures offers a unique advantage. Non-Eurocentric itineraries are vital for advancing and implementing AI across domains, but they can be obstructive and detrimental when they are mere projections from the West. One such prominent technological example is the Mechanical Turk automaton of the eighteenth century, that orientalized the machine, and its contemporary allusive namesake, Amazon Mechanical Turk, that suppresses visual representation of cultural difference (Geoghegan 83, Irani 231). AI will adopt different flavors across cultures, and these nuances are informative both of their eventual lessons and pitfalls. Comparatists can assist in understanding the pivotal, and often underlying, cultural and conceptual choices shaping our future.

While other technological products require a social contract during their widespread adoption, digital technologies do not. Earliest clashes in digital technologies, especially those related to privacy and security, remain unsolved, and generative AI will only

exacerbate them. Technical products, once in the wild, are assessed quantitatively in their societal repercussions with the help of quantitative data science and social sciences, which have thrived in the tech industry in the last decade. Mitigating harmful applications often involves patching a finished product, but the humanities can intervene already at the design step of AI creation. The humanities are tailored for AI, which is much more qualitative than it might appear.

Language has been mastered in the field of natural language processing through highly artificial means of computation—mostly, an entirely distinct mode of language production. Natural language processing and natural language generation are two fields for computational machines, the epitome of the artificial. As Melanie Mitchell wittily notes, in AI speak, natural stands for human (19). Where do humans, as the inventors of these machines, stand when the organismal properties extend to mechanisms?

In addition to computational linguistics, which was designated as a STEM discipline in 2023 (U.S. Immigrations 12), literary and language departments specialize in the study of language. Literature is the space for language in its highest aesthetic form. Machine translation has become indispensable in some translation practices and will affect literary contexts; this work requires not only literal translations but broader translations of cultural contexts and practices. Literary scholars and writers are experimenting with instilling narrative capabilities into large language models. Though the imitation of the human is valuable, the areas where machines diverge from human patterns offer greater creative potential. Machine language is generic and far from beautiful but might unveil previously inaccessible linguistics hinterlands. While we originated all human and computational languages, they are no longer solely ours.

Comparative literature has even more to offer when expanded into theory, particularly in how comparative literature institutionalized critical theory in the U.S. Theoretically and conceptually, in what Ted Underwood calls “the empirical triumph of theory,” large language models are reaching back to Jakobson’s structuralism in language, bolstered by Shannon’s mathematical theory of information. Both were developed in the 1950s and built upon prior work: the first from de Saussure, the second from Markov, who developed his probability method on Pushkin’s poetry. When this probability is implemented in an artificial neural network called a transformer, the human subject becomes irrelevant to the concept of language. A neural architecture trained on text—a hyper-structure trained on a colossal amount of data—is all you need.

Writers like Italo Calvino (1986) and Andrej Tomažin (2018) have picked up on literary connotations of what was then purely fabricated technology before scholars, demonstrating how language machines will contest and re-establish the ideas of Barthes and Foucault, Borges and Oulipo. The value of the humanities should not be in question when esteemed computer scientists are writing and co-authoring papers with literary scholars on topics that used to be exclusive to our domain, from Borges (Bottou and Schölkopf) to narrative theory (Piper, So, and Bamman).

Nowhere in this essay do I argue that AI should be welcomed; instead, I argue that it cannot be stopped. If we want a better future, we better act on it—comparatively. We have the sources, expertise, and depth that cannot be found elsewhere, yet essential to various AI-adjacent domains. We may not have perfected methodologies for such interdisciplinary work, but we can strive to develop them as we address these challenges—collaboratively.

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